Problem 1 (8 pts)

Let c be a real constant, and let

$$A = \begin{bmatrix} 4 & -2 \\ -3 & 5 \\ 6 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 7 & 3 & -4 \\ 2 & c & 5 \end{bmatrix}.$$

a)

(4 pts) Give the shape of each expression (e.g. 5×3), or write "undefined."

$$AB \qquad BA \qquad A + B \qquad A(BA)$$

Solution: The shape of A is 3×2, and the shape of B is 2×3. Therefore,

$$\boxed{AB:\ 3 \times 3, \qquad BA:\ 2 \times 2, \qquad A + B:\ \text{undefined}, \qquad A(BA):\ 3 \times 2.}$$

For a product, the inner dimensions must agree, and the outer dimensions give its shape. Addition requires identical shapes.

b)

(4 pts) Suppose the second column of AB is

$$\begin{bmatrix} 20 \\ -29 \\ 14 \end{bmatrix}.$$

Find the value of c.

Solution: The second column of AB is A times the second column of B:

$$A \begin{bmatrix} 3 \\ c \end{bmatrix} = 3 \begin{bmatrix} 4 \\ -3 \\ 6 \end{bmatrix} + c \begin{bmatrix} -2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 12 - 2c \\ -9 + 5c \\ 18 + c \end{bmatrix}.$$

The first entry gives $12 - 2c = 20$, so $\boxed{c = -4}$. This also gives $-9 + 5(-4) = -29$ and $18 + (-4) = 14$, as required.

Problem 2 (10 pts)

a)

(6 pts) Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation, and suppose

$$T\left(\begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ -4 \end{bmatrix}, \qquad T\left(\begin{bmatrix} -3 \\ 5 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} -5 \\ 7 \end{bmatrix}.$$

i) (1 pt) What is the shape of the matrix corresponding to T, i.e., of the matrix A where $T(\vec{x}) = A\vec{x}$?
Shape:

ii) (5 pts) Find $T\left(\begin{bmatrix} -4 \\ 9 \\ 8 \end{bmatrix}\right)$ without finding the matrix of T.

Solution: i) The domain is $\mathbb{R}^3$ and the codomain is $\mathbb{R}^2$, so the matrix has shape $\boxed{2 \times 3}$.
ii) Observe that

$$\begin{bmatrix} -4 \\ 9 \\ 8 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + 2\begin{bmatrix} -3 \\ 5 \\ 2 \end{bmatrix}.$$

Using the fact that T is a linear transformation,

$$T\left(\begin{bmatrix} -4 \\ 9 \\ 8 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ -4 \end{bmatrix} + 2\begin{bmatrix} -5 \\ 7 \end{bmatrix} = \boxed{\begin{bmatrix} -4 \\ 10 \end{bmatrix}}.$$

b)

(4 pts) Which of the following transformations from $\mathbb{R}^2$ to $\mathbb{R}^2$ are linear? **Select all that apply.**

☐ Rotate 30° clockwise about the origin, then add 1 to the horizontal component.

☐ Reflect across the line $y = -2x$, then multiply the vertical component by 4.

☐ Project orthogonally onto the line $y = 3x$.

☐ Replace the vertical component by its square, leaving the horizontal component unchanged.

Solution: Select the second and third transformations.

- The first is not linear: it sends $\vec{0}$ to $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.
- The second is linear: reflection across a line through the origin is linear, and multiplying the vertical component by 4 is multiplication by $\begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$. A composition of linear transformations is linear.
- The third is linear: projection onto a line through the origin is a linear transformation.
- The fourth is not linear. If $R\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x \\ y^2 \end{bmatrix}$, then $R\left(2\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$, whereas $2R\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$.

Problem 3 (10 pts)

a)

(6 pts) Find the 2×2 matrix P that projects each vector in $\mathbb{R}^2$ onto the line spanned by $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

Solution: Let $\vec{v} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$. The projection of $\vec{x}$ onto its span is

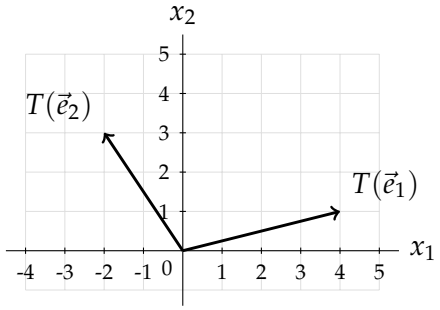
$$\frac{\vec{v} \cdot \vec{x}}{\vec{v} \cdot \vec{v}} \vec{v} = \frac{2x_1 + 5x_2}{29} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \frac{1}{29} \begin{bmatrix} 4x_1 + 10x_2 \\ 10x_1 + 25x_2 \end{bmatrix}.$$

Thus

$$\boxed{P = \frac{1}{29} \begin{bmatrix} 4 & 10 \\ 10 & 25 \end{bmatrix}.}$$

b)

(4 pts) The diagram shows $T(\vec{e}_1)$ and $T(\vec{e}_2)$ for a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$, where $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Find the matrix B corresponding to T.



Solution: The diagram gives $T(\vec{e}_1) = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $T(\vec{e}_2) = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$. These are the first and second columns of the matrix, respectively, so

$$\boxed{B = \begin{bmatrix} 4 & -2 \\ 1 & 3 \end{bmatrix}.}$$

Problem 4 (13 pts)

a)

(4 pts) An invertible 3×3 matrix A satisfies

$$A\begin{bmatrix}4\\-3\\2\end{bmatrix} = \begin{bmatrix}1\\0\\0\end{bmatrix}, \qquad A\begin{bmatrix}-2\\5\\1\end{bmatrix} = \begin{bmatrix}0\\1\\0\end{bmatrix}, \qquad A\begin{bmatrix}3\\1\\4\end{bmatrix} = \begin{bmatrix}0\\0\\1\end{bmatrix}.$$

Find A^{-1} without first finding A. Explain how you found your answer.

Solution: Multiplying each given equation by A^{-1} tells us where the matrix A^{-1} sends the standard basis vectors $\vec{e}_1, \vec{e}_2$, and $\vec{e}_3$. For example, $A^{-1}\vec{e}_1 = \begin{bmatrix}4\\-3\\2\end{bmatrix}$.
These three outputs are the columns of A^{-1}!

$$\boxed{A^{-1} = \begin{bmatrix}4 & -2 & 3\\-3 & 5 & 1\\2 & 1 & 4\end{bmatrix}.}$$

Equivalently, multiplying A by this matrix produces I_3, since the three resulting columns are $\vec{e}_1, \vec{e}_2, \vec{e}_3$.

b)

(4 pts) Find B^{-1}, where

$$B = \begin{bmatrix}5 & 4\\6 & 8\end{bmatrix}.$$

Solution: Using the two-by-two inverse formula,

$$B^{-1} = \frac{1}{5(8)-4(6)}\begin{bmatrix}8 & -4\\-6 & 5\end{bmatrix} = \boxed{\frac{1}{16}\begin{bmatrix}8 & -4\\-6 & 5\end{bmatrix}.}$$

The denominator is nonzero, so the inverse exists.

c)

(5 pts) An invertible matrix C has inverse

$$C^{-1} = \begin{bmatrix} 3 & -2 \\ 4 & -1 \end{bmatrix}.$$

Solve for $\vec{x}$:

$$C\left(4\vec{x} + \begin{bmatrix} 5 \\ -4 \end{bmatrix}\right) = \begin{bmatrix} 7 \\ 4 \end{bmatrix}.$$

Solution: Multiply both sides by C^{-1}:

$$4\vec{x} + \begin{bmatrix} 5 \\ -4 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 7 \\ 4 \end{bmatrix} = \begin{bmatrix} 13 \\ 24 \end{bmatrix}.$$

Consequently,

$$4\vec{x} = \begin{bmatrix} 8 \\ 28 \end{bmatrix}, \qquad \boxed{\vec{x} = \begin{bmatrix} 2 \\ 7 \end{bmatrix}.}$$

Problem 5 (16 pts)

a)

(8 pts) Find a scalar k and matrices P, L, and U such that $PA = LU$, where

$$A = \begin{bmatrix} 0 & 3 & -4 \\ 4 & -2 & 6 \\ -4 & 8 & k \end{bmatrix}, \qquad U = \begin{bmatrix} 4 & a & b \\ 0 & 3 & c \\ 0 & 0 & 5 \end{bmatrix}.$$

Here, P is a permutation matrix, and L is lower triangular with ones on its diagonal. Write all entries of P, L, and U in the labeled boxes.

Solution: The first row of LU equals the first row of U. Its first entry is 4, so put the second row of A first. Choose

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad PA = \begin{bmatrix} 4 & -2 & 6 \\ 0 & 3 & -4 \\ -4 & 8 & k \end{bmatrix}.$$

Write

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}.$$

Matching the first row of LU with PA gives $a = -2$ and $b = 6$. Matching the first column gives

$$4\ell_{21} = 0, \qquad 4\ell_{31} = -4, \qquad \text{so} \qquad \ell_{21} = 0, \quad \ell_{31} = -1.$$

Entry $(2,3)$ gives $0(6) + c = -4$, so $c = -4$. Entry $(3,2)$ gives

$$(-1)(-2) + 3\ell_{32} = 8, \qquad \text{so} \qquad \ell_{32} = 2.$$

Finally, entry $(3,3)$ gives $k = (-1)(6) + 2(-4) + 5 = -9$. Therefore,

$$\boxed{k = -9, \qquad P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},}$$

$$\boxed{L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} 4 & -2 & 6 \\ 0 & 3 & -4 \\ 0 & 0 & 5 \end{bmatrix}.}$$

As a check, $LU = \begin{bmatrix} 4 & -2 & 6 \\ 0 & 3 & -4 \\ -4 & 8 & -9 \end{bmatrix} = PA$.

b)

(8 pts) A different 3×3 matrix B has the factorization $PB = LU$, where

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \qquad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -2 & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} 3 & 4 & -2 \\ 0 & -5 & 5 \\ 0 & 0 & 4 \end{bmatrix}.$$

Use these factors to solve

$$B\vec{x} = \begin{bmatrix} 26 \\ 8 \\ 1 \end{bmatrix}.$$

For full credit, solve using two triangular systems; don't find the entries of B directly.

Solution: First reorder the right-hand side:

$$P \begin{bmatrix} 26 \\ 8 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \\ 26 \end{bmatrix}.$$

Solve $L\vec{y} = P\vec{b}$ by forward substitution:

$$y_1 = 8, \qquad 2y_1 + y_2 = 1, \qquad -y_1 - 2y_2 + y_3 = 26.$$

Thus $y_1 = 8$, $y_2 = -15$, and $y_3 = 4$. Next, solve $U\vec{x} = \vec{y}$ by back substitution:

$$\begin{aligned} 4x_3 = 4 \quad &\Longrightarrow \quad x_3 = 1, \\ -5x_2 + 5x_3 = -15 \quad &\Longrightarrow \quad x_2 = 4, \\ 3x_1 + 4x_2 - 2x_3 = 8 \quad &\Longrightarrow \quad x_1 = -2. \end{aligned}$$

Therefore $\boxed{\vec{x} = \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}}$. The entries are already in the original variable order.

Problem 6 (16 pts)

a)

(8 pts) Let

$$A = \begin{bmatrix} 2 & -4 & 6 & 0 \\ -4 & 8 & -9 & 6 \\ 6 & -12 & 15 & -6 \end{bmatrix}.$$

Find all real numbers s for which the following system has infinitely many solutions:

$$A\vec{x} = \begin{bmatrix} s \\ 1 \\ 4 \end{bmatrix}.$$

If there are multiple such values of s, separate them with commas. You do not need to write the solution set.

Solution: Find an LU factorization by matching entries in the product. The first row of U is the first row of A. Matching the first column gives $\ell_{21} = -2$ and $\ell_{31} = 3$. The second row must then satisfy

$$(-2)(-4) + u_{22} = 8, \qquad (-2)(6) + u_{23} = -9, \qquad u_{24} = 6,$$

so its entries are $0, 0, 3, 6$. Matching entry $(3,3)$ gives $3(6) + 3\ell_{32} = 15$, so $\ell_{32} = -1$. The remaining entries of the last row of U are zero. Hence

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & -4 & 6 & 0 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{U}.$$

Solve $L\vec{y} = \begin{bmatrix} s \\ 1 \\ 4 \end{bmatrix}$:

$$y_1 = s, \qquad -2y_1 + y_2 = 1, \qquad 3y_1 - y_2 + y_3 = 4.$$

This gives $\vec{y} = \begin{bmatrix} s \\ 1 + 2s \\ 5 - s \end{bmatrix}$. The last equation of $U\vec{x} = \vec{y}$ is $0 = 5 - s$, so consistency requires $s = 5$.

When $s = 5$, the zero row gives $0 = 0$. Columns 1 and 3 contain pivots, and x_2, x_4 are free: the second equation determines x_3 and the first determines x_1 for every choice of x_2, x_4. Thus there are infinitely many solutions exactly when $\boxed{s = 5}$.

Problem 6(a): Direct solution

Solution: Another approach is to try to solve directly. The three equations are

$$\begin{aligned} 2x_1 - 4x_2 + 6x_3 &= s, \\ -4x_1 + 8x_2 - 9x_3 + 6x_4 &= 1, \\ 6x_1 - 12x_2 + 15x_3 - 6x_4 &= 4. \end{aligned}$$

Solve the first equation for x_1:

$$x_1 = \frac{s}{2} + 2x_2 - 3x_3.$$

Substituting this into the second equation gives

$$\begin{aligned} -4\left(\frac{s}{2} + 2x_2 - 3x_3\right) + 8x_2 - 9x_3 + 6x_4 &= 1, \\ -2s + 3x_3 + 6x_4 &= 1, \\ 3x_3 + 6x_4 &= 1 + 2s. \end{aligned}$$

Substituting the same expression for x_1 into the third equation gives

$$\begin{aligned} 6\left(\frac{s}{2} + 2x_2 - 3x_3\right) - 12x_2 + 15x_3 - 6x_4 &= 4, \\ 3s - 3x_3 - 6x_4 &= 4, \\ 3x_3 + 6x_4 &= 3s - 4. \end{aligned}$$

These two equations have the same left-hand side, so they are only consistent if

$$1 + 2s = 3s - 4 \qquad \Longrightarrow \qquad \boxed{s = 5}.$$

When $s = 5$, both equations become $3x_3 + 6x_4 = 11$. We can choose $x_2 = t$ and $x_4 = u$ freely, then obtain

$$x_3 = \frac{11}{3} - 2u, \qquad x_1 = \frac{5}{2} + 2t - 3\left(\frac{11}{3} - 2u\right) = -\frac{17}{2} + 2t + 6u.$$

Every choice of $t, u \in \mathbb{R}$ gives a solution of all three original equations. Thus there are infinitely many solutions when $s = 5$, and no solutions for any other value of s.

b)

(8 pts) Find all solutions of the following system. Give your answer in vector-parametric form. Circle your final answer.

$$\begin{bmatrix} 2 & 0 & -4 & 6 & 0 \\ 0 & 3 & 6 & -3 & 0 \\ 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 8 \\ -9 \\ 20 \\ 0 \end{bmatrix}.$$

Solution: The equations are

$$2x_1 - 4x_3 + 6x_4 = 8, \qquad 3x_2 + 6x_3 - 3x_4 = -9, \qquad 5x_5 = 20, \qquad 0 = 0.$$

The pivot variables are x_1, x_2, x_5, and the free variables are x_3, x_4. Let $x_3 = s$ and $x_4 = t$. Then

$$x_1 = 4 + 2s - 3t, \qquad x_2 = -3 - 2s + t, \qquad x_5 = 4.$$

The complete solution set is

$$\boxed{\vec{x} = \begin{bmatrix} 4 \\ -3 \\ 0 \\ 0 \\ 4 \end{bmatrix} + s \begin{bmatrix} 2 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \qquad s, t \in \mathbb{R}.}$$

Problem 7 (11 pts)

a)

(5 pts) Let

$$S = \left\{ \begin{bmatrix} 3a - 2b \\ 6a - 4b \\ 5a + b \end{bmatrix} : a, b \in \mathbb{R} \right\}.$$

Find a 3×3 matrix A such that $\text{colsp}(A) = S$.

Solution: Write each vector in S as

$$\begin{bmatrix} 3a - 2b \\ 6a - 4b \\ 5a + b \end{bmatrix} = a \begin{bmatrix} 3 \\ 6 \\ 5 \end{bmatrix} + b \begin{bmatrix} -2 \\ -4 \\ 1 \end{bmatrix}.$$

Use these two vectors as the first two columns of A. To get the required 3×3 shape, the third column should be any other vector in S: any linear combination of these first two columns. An easy choice is $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, obtained by taking both coefficients to be zero. Thus one answer is

$$\boxed{A = \begin{bmatrix} 3 & -2 & 0 \\ 6 & -4 & 0 \\ 5 & 1 & 0 \end{bmatrix}.}$$

Every linear combination of its columns belongs to S, and every vector of S is obtained using coefficients $a, b, 0$. The zero column does not change the column space.

b)

(6 pts) For each set below, explain why it is **not** a subspace of $\mathbb{R}^3$.

i) (2 pts)

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : z = 1 \right\}.$$

ii) (4 pts)

$$V = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : y = 0,\ z = x^2 \right\}.$$

Solution: i) The zero vector does not belong to W, because its third entry is 0, not 1. Every subspace contains the zero vector, so W is not a subspace.

ii) The vector $\vec{v} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ belongs to V, since $1 = 1^2$. However, $2\vec{v} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$ does not belong to V, since $2 \neq 2^2$. Thus V is not closed under scalar multiplication and is not a subspace.

Problem 8 (16 pts)

a)

(8 pts) Let $\vec{p}, \vec{q}, \vec{r} \in \mathbb{R}^7$ be linearly independent. Find the rank of each matrix.

$$A = \begin{bmatrix} | & | & | & | \\ \vec{p}+\vec{q} & \vec{q}+\vec{r} & \vec{p}+\vec{r} & 4\vec{r} \\ | & | & | & | \end{bmatrix}$$

$$B = \begin{bmatrix} | & | \\ 2\vec{p}-\vec{q} & 8\vec{p}-4\vec{q} \\ | & | \end{bmatrix}$$

$$C = \begin{bmatrix} | & | & | \\ \vec{p}+2\vec{q} & 3\vec{q}-\vec{r} & 2\vec{p}+7\vec{q}-\vec{r} \\ | & | & | \end{bmatrix}$$

$$D = \begin{bmatrix} | & | & | & | \\ \vec{p}+\vec{q} & \vec{q}+\vec{r} & \vec{p}-\vec{r} & 2\vec{p}+\vec{q}-\vec{r} \\ | & | & | & | \end{bmatrix}.$$

Solution: The rank is the dimension of the column space.
A**: rank 3.** All four columns lie in $\text{span}(\vec{p}, \vec{q}, \vec{r})$, so the rank is at most 3. Conversely, writing its columns as $\vec{a}_1, \ldots, \vec{a}_4$,

$$\vec{r} = \tfrac{1}{4}\vec{a}_4, \qquad \vec{q} = \vec{a}_2 - \tfrac{1}{4}\vec{a}_4, \qquad \vec{p} = \vec{a}_3 - \tfrac{1}{4}\vec{a}_4.$$

Its column space therefore contains the three independent vectors $\vec{p}, \vec{q}, \vec{r}$, giving $\boxed{\text{rank}(A) = 3}$.
B**: rank 1.** Its second column is four times its first. The first column is nonzero, since $2\vec{p} - \vec{q} = \vec{0}$ would contradict the independence of $\vec{p}, \vec{q}, \vec{r}$. Thus $\boxed{\text{rank}(B) = 1}$.
C**: rank 2.** Its third column is twice its first plus its second. The first two are independent: if

$$\alpha(\vec{p} + 2\vec{q}) + \beta(3\vec{q} - \vec{r}) = \vec{0},$$

then $\alpha\vec{p} + (2\alpha + 3\beta)\vec{q} - \beta\vec{r} = \vec{0}$. Independence of $\vec{p}, \vec{q}, \vec{r}$ forces $\alpha = \beta = 0$. Hence $\boxed{\text{rank}(C) = 2}$.
D**: rank 2.** Let $\vec{d}_1 = \vec{p} + \vec{q}$ and $\vec{d}_2 = \vec{q} + \vec{r}$. The remaining columns are $\vec{d}_1 - \vec{d}_2$ and $2\vec{d}_1 - \vec{d}_2$. Moreover,

$$\alpha\vec{d}_1 + \beta\vec{d}_2 = \alpha\vec{p} + (\alpha + \beta)\vec{q} + \beta\vec{r} = \vec{0}$$

forces $\alpha = \beta = 0$. Thus $\vec{d}_1, \vec{d}_2$ are a basis for the column space, and $\boxed{\text{rank}(D) = 2}$.

b)

(2 pts) A 6×8 matrix E has rank 3. How many free variables does $E\vec{x} = \vec{0}$ have?

Number of free variables:

Solution: Rank 3 means that an echelon factor U of E has 3 pivots. For example, one possible arrangement is

$$U = \begin{bmatrix} \boxed{p_1} & * & * & * & * & * & * & * \\ 0 & 0 & \boxed{p_2} & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \boxed{p_3} & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \qquad p_1, p_2, p_3 \neq 0.$$

The boxed entries are the pivots; each $*$ can be any number. In this example, the variables are grouped as

Variable	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
Type	pivot	free	pivot	free	free	pivot	free	free

The actual pivot columns may differ, but there must be 3 pivot columns among the 8 columns. The other $\boxed{8 - 3 = 5}$ columns give free variables. The homogeneous system is always consistent, so all five free variables can be chosen freely.

c)

(3 pts) A 4×7 matrix F has rank 4. Is every vector in $\mathbb{R}^4$ a possible output of F? Briefly explain.

Solution: Yes. The column space of F is a subspace of $\mathbb{R}^4$ with dimension 4, so it equals $\mathbb{R}^4$. Thus every vector in $\mathbb{R}^4$ is a possible output of F.

d)

(3 pts) A 6×4 matrix G has rank 4. Could $G\vec{x} = \vec{b}$ have exactly one solution for some $\vec{b} \in \mathbb{R}^6$? Briefly explain.

Solution: Yes. Since G has rank 4 and 4 columns, its echelon factor U has a pivot in every column. There are no free variables, so any consistent system has exactly one solution, determined by back substitution.
For example, choose $\vec{b} = \vec{0}$. This system is consistent because $\vec{x} = \vec{0}$ is a solution. Since there are no free variables, this solution is unique.